

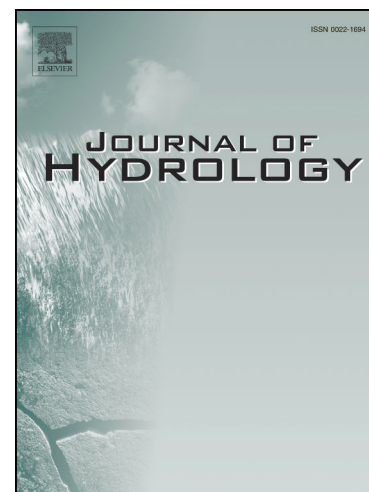
Spatial-heterogeneity-aware deep learning for water quality prediction: a case study in Hong Kong

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Keyword: Water quality prediction; Spatial heterogeneity; Spatially adaptive regionalization; Deep learning; Explainable artificial intelligence.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

1. Four homogeneous water-quality subregions are delineated by adaptive regionalization.
2. Regional prediction models outperform global models across all window lengths.
3. The optimal model is identified as a 6-month-window LSTM-TCN-LSTM-LSTM architecture.
4. Historical WQI is the most vital input, with a strong positive effect on predictions.

Spatial-heterogeneity-aware deep learning for water quality prediction: A case study in Hong Kong

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1. Introduction

River water pollution has emerged as a pressing environmental challenge, primarily driven by a complex interplay of anthropogenic activities, such as industrial effluent, domestic wastewater, and agricultural runoff (Prasad et al., 2022; Mummidivarapu et al., 2023; Pei et al., 2024). The impacts of these driving forces vary over time, as the type and intensity of human activities fluctuate across seasons and socio-economic development stages, thereby causing dynamic changes in water quality (Liu et al., 2025; Pak et al., 2021; Wu and Lu, 2021; Yu and Lu, 2018; Sun et al., 2016). The temporal variability nature of water quality underscores the necessity of predicting future water quality conditions. While continuous monitoring provides essential insights into the current state of water bodies, it has limited capacity to anticipate the evolving trajectories of water quality. In contrast, water quality prediction seeks to infer future water quality conditions by analyzing historical datasets and identifying underlying temporal regularities (Wan et al., 2022; Prasad et al., 2022; Wang et al., 2024; Wan et al., 2025). Conducting such predictive efforts is of considerable importance, as it not only facilitates early warning and risk reduction, but also provides scientific support for the formulation and implementation of environmental management policies.

In general, research on water pollution prediction generally falls into two categories: qualitative prediction and quantitative prediction (Cheng et al., 2021; Chong et al., 2023; Chen et al., 2019; Gao et al., 2023; Xie et al., 2024). Qualitative prediction represents a non-numerical, descriptive analytical approach, which emphasizes understanding and interpreting the underlying causes, trends, and potential impacts of future changes in water quality. For instance, Varol (2020a) employed the Mann-Kendall (M-K) test to examine the seasonal changes of water quality parameters (WQPs) in the Keban Reservoir, Turkey, and found that elevated water temperature (WT) reduced dissolved oxygen (DO) and accelerated organic matter decomposition, thereby causing lower DO and higher chemical oxygen demand (COD) in summer than in winter. Similarly, Xiang et al. (2021) pointed out that nutrient pollution in the midstream Three Gorges Reservoir was likely to intensify, as rising total phosphorus (TP) levels aggravated downstream algal blooms, underscoring the need for nutrient treatment and non-point source pollution control in future water quality management. Ghaderpour et al. (2024) applied Least-Squares Cross-Wavelet Analysis in the Sacco River Valley, Italy. They found significant seasonal correlations between precipitation and streamflow, which may have implications for ground deformation and could provide insights for water quality prediction in similar contexts. In Yun et al.'s (2015) study, the seasonal variation trends of WQPs in the mainstream of the Nakdong River Basin, Korea were explored, revealing that while biochemical oxygen demand (BOD) and TP exhibited decreasing trends at most sites, COD and suspended solids (SS) showed increasing trends. Kuwayama et al. (2020) predicted the temporal variation of WQPs in 23 river sub-basins in Texas, USA, and found degrading trends in DO, TN, TP, and chlorophyll-a, while noted that nutrient pollution in the state's surface waters may be an increasingly severe issue. Trend analysis of river water quality provides a rapid diagnostic of change direction, enabling practitioners to identify potential issues and implement timely interventions to prevent deterioration. However, such analysis offers only qualitative judgments (e.g., improvement vs. degradation) and does not quantify the magnitude

of change, limiting its utility for precise water quality early-warning applications and operational decision-making.

To quantitatively characterize the future state of river water quality, recent studies have increasingly leveraged deep learning and other artificial intelligence techniques to capture and represent the intrinsic features of water quality data (Bi et al., 2023; Tian et al., 2024). Deep learning methods have demonstrated strong performance in predicting future water quality because they can effectively model nonlinear dynamics and complex temporal variations in water quality time series (Guo et al., 2023; You et al., 2024; Meng et al., 2025). For example, Talukdar et al. (2023) proposed a deep-learning-based stacked ensemble model that achieved accurate prediction of the water quality index (WQI) of Loktak Lake in India and, through sensitivity and uncertainty analyses, identifies key drivers affecting prediction performance. Peng et al. (2022) introduced a Transformer-based deep transfer learning framework that improved long-term prediction accuracy for multiple parameters, with particular gains at data-sparse monitoring sites, offering a novel technical pathway for long-term forecasting and decision support. In another study, Zheng et al. (2023) applied an interpretable deep learning model to quantitatively predict the spatiotemporal variations of stream WQPs across a broad region of southern China. More recently, Wan et al. (2025) developed a deep learning model, spatiotemporal feature graph neural network for multi-point water quality prediction, which effectively captured the spatiotemporal dependencies in river networks and further enhanced prediction performance. Wang et al. (2025) proposed a spatiotemporal prediction method that integrated deep learning with spatial interpolation, achieving superior accuracy for DO and total nitrogen relative to traditional methods. Recently, a deep-learning-based Restricted Boltzmann Machine (Xu et al., 2022), which can capture complex temporal patterns and reduce dimensionality, has been used to predict water portability from water quality data (Rao et al., 2024).

Although machine learning and deep learning techniques have been widely recognized as effective tools for improving water quality prediction (Nong et al., 2025; Yu et al., 2022; Xu et al., 2024), most existing studies assume uniform water quality variation within the study area and construct global prediction models. In reality, river water quality exhibits strong spatial heterogeneity (Fotheringham et al., 2017; Goodchild et al., 2021) because subregions may differ in pollution sources, land use, hydrological conditions, and human activities. As a result, different subregions may exhibit distinct temporal patterns and may be influenced by different key drivers, such as domestic wastewater, agricultural runoff, and industrial effluents. Treating the entire study area uniformly may limit the ability of models to capture regional temporal patterns. This can make it difficult to accurately predict water quality changes in specific subregions and weaken the effectiveness of regional water quality management and early warning.

To address this problem, this study develops a spatial-heterogeneity-aware water quality prediction framework that captures the local dynamics of water quality and its key drivers. This framework advances existing research by (1) introducing spatially adaptive clustering to delineate subregions with relatively homogeneous water quality, enabling a more rational subregional division that supports identification of subregion-specific key water quality

parameters and better captures local water quality dynamics; (2) systematically evaluating both global and subregion-specific deep learning models under varying sliding-window lengths to improve the accuracy of water quality predictions; (3) leveraging interpretable artificial intelligence, SHapley Additive exPlanations (SHAP), to quantify the contributions of key input parameters and improve understanding of spatially heterogeneous drivers of water quality. A case study using water quality monitoring data in Hong Kong demonstrates the effectiveness of the proposed framework in capturing local water quality dynamics and providing insights for region-specific water quality management and early warning.

2. Material and Methods

Fig. 1 shows the proposed framework for water quality prediction, including the data used, the methods, and the outputs. Each of these components is described in the following.

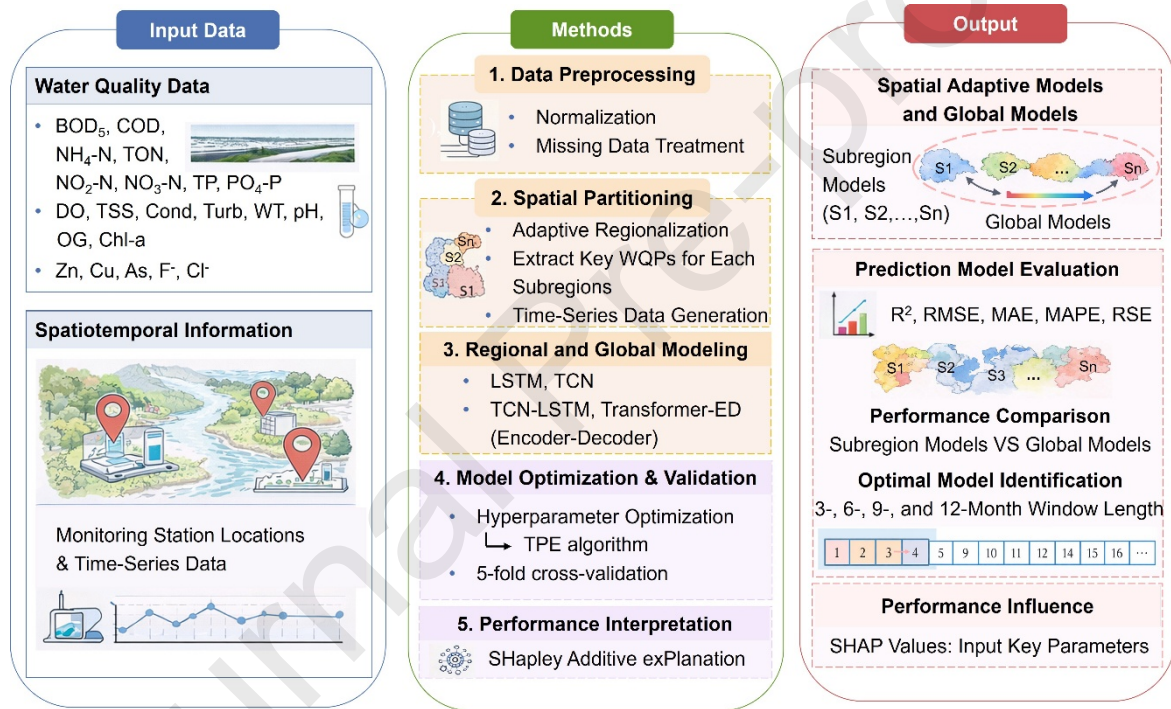


Fig. 1. The proposed framework for water quality prediction

2.1. Study area and data

The study area is situated in the east-central part of Hong Kong, which is characterized by a subtropical monsoon climate with an annual average temperature of about 23.3 °C. Hong Kong's surface water system is primarily composed of rivers, streams and nullahs (HKEPD, 2021). For convenience, the aforementioned surface waters are collectively referred to as river water. In the 1980s, rivers in Hong Kong experienced severe pollution from livestock farming, domestic sewage, and industrial wastewater. While the water quality situation gradually improved following the implementation of various pollution control measures, the issue of water pollution persists, and continuous water quality monitoring is conducted.

This study focuses on four water control zones, including Tolo Harbour and Channel, Victoria Harbour, Junk Bay, and Port Shelter, with a total of 42 monitoring stations (<https://www.epd.gov.hk/>) (Fig. 2). Monthly measurements were conducted at each station, providing a continuous and systematic record of various WQPs for long-term analysis. The dataset consists of water quality monitoring records collected over a ten-year period (January 2010-December 2019). In total, 21 commonly used WQPs are considered for water quality analysis, including 5-day biochemical oxygen demand (BOD_5), chemical oxygen demand (COD), dissolved oxygen (DO), ammonium nitrogen (NH_4-N), total Kjeldahl nitrogen (TON), nitrite-nitrogen (NO_2-N), nitrate-nitrogen (NO_3-N), total phosphorus (TP), orthophosphate phosphorus (PO_4-P), total suspended solids (TSS), conductivity (Cond), turbidity (Turb), water temperature (WT), potential of hydrogen (pH), oil and grease (OG), chlorophyll-a (Chl-a), fluoride (F^-), chloride (Cl^-), zinc (Zn), arsenic (As), and copper (Cu). These monitoring data provide a robust basis for examining spatiotemporal variations in river water quality across the four water control zones.

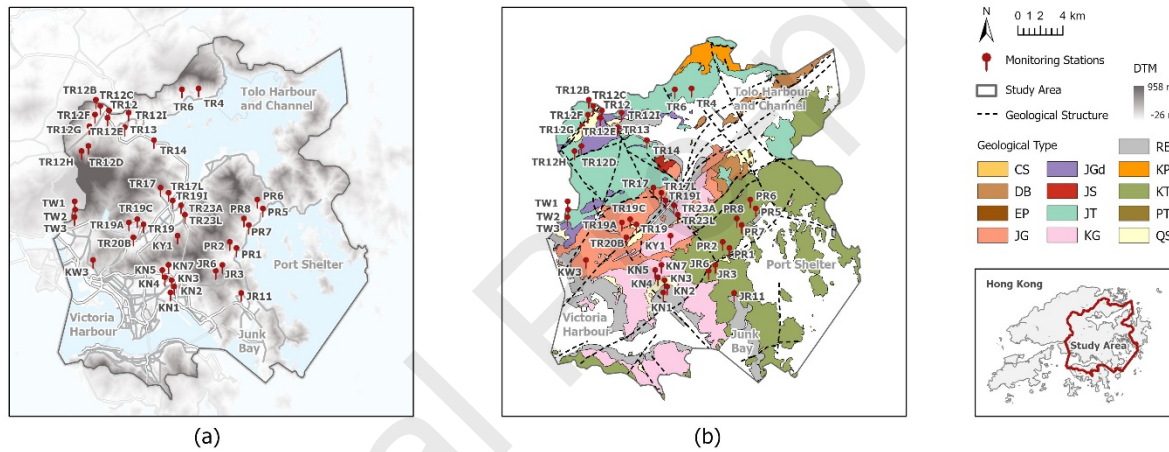


Fig. 2. Spatial distribution of monitoring stations within the study area over base maps: (a) digital terrain model; (b) geological types.

2.2. Method description

2.2.1. Water quality index

The water quality index (WQI) serves as a comprehensive quantitative tool that integrates multiple WQPs into a unified numerical representation, thereby offering an efficient approach to interpreting and assessing the overall condition of surface water quality (Wu et al., 2018; Pan et al., 2022; Zhang et al., 2024). In this study, water quality is assessed using the classical WQI calculation method proposed by Pesce and Wunderlin (2000). This method has been widely applied in water quality assessment studies and adopted in various monitoring programs, demonstrating high reliability and broad applicability (Xiao et al., 2025; Wu et al., 2024; Varol, 2020; Qu et al., 2020; El Najjar et al., 2019). The calculation formula is expressed as follows:

$$WQI = \sum_{i=1}^n C_i P_i / \sum_{i=1}^n P_i \quad (1)$$

where C_i represents the normalized value of parameter i , n represents the total number of selected parameters, and P_i represents the relative weight assigned to parameter i . The specific values of C_i and P_i are presented in Table S1, and the criteria for water quality classification based on WQI are provided in the Supplementary Material (Section S1).

2.2.2. Spatially adaptive regionalization

This study employs the regionalization with dynamically constrained agglomerative clustering and partitioning (REDCAP) method to identify relatively homogeneous subregions of river water quality. The subregions obtained through clustering and regionalization are subsequently used as spatial units for analyzing regional variations in water quality. The REDCAP algorithm consists of two primary steps (Guo, 2008; Kupfer et al., 2012, Boluwade, 2019): (1) constructing a spatially contiguous tree using an adjacency-constrained clustering method; and (2) partitioning this tree into spatial subtrees, each representing a contiguous spatial region. At every hierarchical level, REDCAP enforces spatial contiguity for all clusters. Contiguity is maintained through the clustering and optimization processes, yielding more coherent and efficient derived regions. The method iteratively merges the most similar spatially connected units to build the spatially contiguous tree. The full-order constrained average linkage clustering (Full-Order-ALK) is used to perform clustering and partitioning so as to delineate homogeneous regions, as it exhibits superior performance on four quality criteria: overall heterogeneity, within-region variability, regional size balance, and preservation of the data distribution (Guo, 2008). The full computational formulas and detailed descriptions about the REDCAP algorithm are provided in the Supplementary Material (Section S2).

2.2.3. Deep learning methods

River water quality exhibits complex and diverse temporal dynamics due to varying pollution sources, hydrological conditions, and human activities. The water quality in different subregions may exhibit distinct short-term fluctuations, seasonal cycles, and long-term trends, making it difficult to capture all relevant patterns with a single model. To address this, this study evaluates four representative deep learning methods for water quality prediction, including temporal convolutional network (TCN), long short-term memory (LSTM), a hybrid TCN-LSTM encoder-decoder, and a Transformer-based encoder-decoder (Transformer-ED). Each model has complementary strengths in modeling temporal dependencies, and evaluating these models allows the study to identify the most suitable architecture for each scenario under both global and regional settings.

These methods differ in their mechanisms for handling temporal dependencies, parallelization, and long-range sequence modeling. The TCN leverages dilated causal convolutions and residual connections to efficiently model long-range dependencies, whereas the LSTM introduces gating mechanisms to retain and update information across extended sequences. The hybrid TCN-LSTM combines the receptive-field expansion of TCN with the

memory control of LSTM within an encoder–decoder framework, while the Transformer-ED exploits self-attention and multi-head attention to directly capture global dependencies in parallel. Given the complexity of these architectures, their mathematical formulations, structural diagrams, and operational principles are provided in the Supplementary Material (Section S3).

2.3. Modeling of water quality prediction

2.3.1. Global and regional prediction models

To investigate the performance differences between global and regional modeling approaches for river water quality prediction, this study employs the deep learning algorithms mentioned before to construct intelligent prediction models under both global and regional scenarios.

Global modeling scenario: Global water quality prediction models are developed using four deep learning algorithms—TCN, LSTM, TCN-LSTM, and Transformer-ED under sliding time windows of 3, 6, 9, and 12 months.

Regional modeling scenario: Regional prediction models are constructed within spatial subregions employing the same four deep learning algorithms with sliding-window lengths of 3, 6, 9, and 12 months.

This study aims to forecast future WQI values based on historical water quality patterns. Therefore, the prediction models include historical WQI as an input. It is necessary because historical WQI observations directly reflect recent water quality trends and temporal dependencies. Using historical WQI does not cause information leakage because the model only uses observations from previous time steps to predict future values. Moreover, the models do not rely solely on past observations as in simple autoregressive smoothing. Other water quality parameters are also included to account for additional factors influencing future WQI, allowing the models to capture complex relationships and improve forecasting accuracy. The models enable timely forecasting of future water quality conditions based on current and past observations of water quality parameters and WQI, providing actionable information for water management.

The modeling process of the river water quality prediction primarily consists of four steps. First, water quality monitoring data in the study area are normalized to the range [0, 1] while preserving the data distribution, thus reducing the impact of varying magnitudes on prediction performance. Second, at the global level, correlation analysis and stepwise regression are employed to identify key global input parameters for water quality prediction. Global prediction models for water quality are developed based on the TCN, LSTM, TCN-LSTM, and Transformer-ED algorithms under time window lengths of 3, 6, 9, and 12 months, respectively. The prediction accuracy of these global models is then compared. Third, at the regional level, the REDCAP method is applied to perform spatially adaptive clustering and partitioning of the study area, thereby delineating spatially homogeneous water quality subregions. Within each subregion, correlation analysis and stepwise regression are again used to identify key regional

input parameters. Following this, regional models for water quality prediction are established based on TCN, LSTM, TCN-LSTM, and Transformer-ED algorithms under time window lengths of 3, 6, 9, and 12 months, respectively. The performance differences among these models are subsequently evaluated to determine their suitability for regional water quality prediction.

2.3.2. Time series sample generation

To ensure the effective evaluation of water quality prediction models and to facilitate the optimal model selection, all models are trained and tested under identical experimental settings. The training dataset comprises monitoring data from 2010 to 2017, while data from 2018 to 2019 served as the test dataset. Furthermore, five-fold cross-validation is performed on the training dataset by randomly partitioning it into five subsets of approximately equal size.

When applying deep learning methods to predict future water quality states, the collected and normalized water quality data cannot be directly employed in model training. Instead, these data must be transformed into model-recognizable samples and labels. To achieve this, a sliding time window approach is utilized to convert continuous time series into new input samples suitable for training and testing (Meng, 2019; Bhatt et al., 2022). To predict the value at time step t , we use the values at time steps $t-1, t-2, \dots, t-\Delta t$ as inputs. Here, Δt represents the length of the sliding window. This approach preserves temporal dependence, as each training sample always uses past observations to predict a future value, ensuring that information flows in chronological order. Given the seasonal and periodic characteristics of water quality monitoring data, the sliding-window lengths of 3, 6, 9, and 12 months are adopted to capture temporal patterns at different scales. The stride between consecutive windows is set to one time step. The procedure for generating input samples for water quality prediction is depicted in Fig. 3.

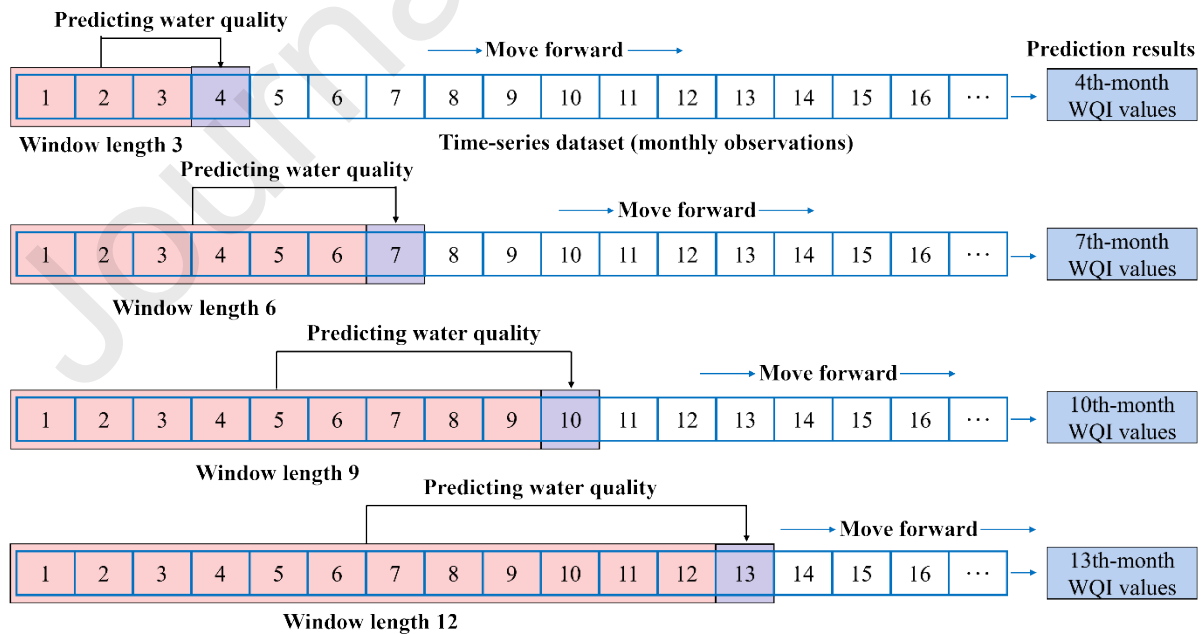


Fig. 3. Schematic representation of one-step-ahead time series construction for water quality prediction under different sliding-window lengths.

2.3.3. Model evaluation

The prediction accuracy of the constructed models is evaluated using five performance metrics: the coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE) and correlation coefficient (CC) (Bui et al., 2020; Zheng et al., 2023; Wang et al., 2025). In addition to these aggregate metrics, pointwise residuals (RES) are analyzed to characterize the distribution and systematic patterns of the prediction errors. Their corresponding formulas are presented as follows:

$$R^2 = \frac{\sum_{i=1}^n (A_i - \bar{A}_i)^2 - \sum_{i=1}^n (A_i - P_i)^2}{\sum_{i=1}^n (A_i - \bar{A}_i)^2} \quad (2)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i - A_i)^2} \quad (3)$$

$$\text{MAE} = \frac{\sum_{i=1}^n |P_i - A_i|}{n} \quad (4)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{P_i - A_i}{A_i} \right| \quad (5)$$

$$\text{RES} = A_i - P_i \quad (6)$$

where P_i and A_i denote the predicted and actual WQI values, respectively, while $\bar{A}_i$ denotes the mean of the actual WQI values. The values of RMSE, MAPE, and MAE fall within the range of 0 to 1, with lower values indicating higher accuracy and greater reliability of the prediction model. The R^2 is an important indicator for evaluating the goodness of fit; the closer its value is to 1, the better the model performance. RES denotes the discrepancies between actual and predicted values, with smaller values suggesting a better model fit and greater prediction accuracy.

3. Results

3.1. Spatial subregions and input parameters for prediction modeling

3.1.1. Spatially adaptive regionalization

The water quality monitoring stations in the study area are adaptively clustered and regionalized to enhance the internal homogeneity of research units by jointly considering the water quality characteristics and geographic locations. The input data include the WQI, representing water quality characteristics, and the spatial coordinates of the monitoring stations. To ensure the robustness and validity of the partitioning, five alternative runs of spatially adaptive regionalization are conducted, each yielding a distinct result. As illustrated in Fig. S5, each color represents a subregion. The initial partition divides the study area into two subregions; the second partition produces three subregions, and the process continues until the fifth partition, which yields six subregions. Considering the overall heterogeneity among subregions, the four-subregion scheme is identified as the optimal configuration, with the lowest SDD value of 933. In Fig. S5c, subregions 1-4 (S1-4) contained 4, 12, 18, and 8 monitoring stations, respectively, which are spatially contiguous and geographical adjacent. Subsequent extraction of key WQPs and the development of regional prediction models are both based on this clustering and regionalization result. By combining spatial contiguity constraints with statistical optimization, the REDCAP algorithm generates subregions that are environmentally meaningful and geographically consistent. These subregions provide an appropriate foundation for analyzing regional variations in river water quality, enabling the detection of spatial heterogeneity that might otherwise be obscured by global analyses.

3.1.2. Input water quality parameters

The input combination of WQPs exerts a significant influence on the performance of water quality prediction models. Following prior work (Choubin et al., 2017; Rajaei et al., 2019; Meng et al., 2021; Xu et al., 2024), correlation analysis is first used to quantify the associations between WQPs and the WQI, and parameters showing strong positive or negative correlations are considered as candidate inputs. To eliminate redundant or less contribution parameters, stepwise regression analysis is then employed to further optimize the parameter set. As shown in Table S3, COD exhibits the strongest correlation with WQI in the global study area as well as in S1-2 and S4, whereas Cond is most strongly correlated with WQI in S3. This parameter-by-parameter analysis supports the sequential selection of appropriate WQPs. When strong correlations exist among the extracted WQPs, stepwise regression can be used to progressively remove those parameters contributing less to the WQI (Table S4). The key WQPs are determined for both the global region and each local subregion based on their correlations with WQI and respective contributions to WQI. For the global region, the selected parameters are COD, NO₂-N, DO, PO₄-P, Cond, Cu, and F⁻. For S1, the key parameters are COD, BOD₅, F⁻, DO, NO₂-N, SS, and As; for S2, BOD₅, COD, SS, NO₂-N, Cond, F⁻, and NH₄-N; for S3, Cond, TKN, Cu, F⁻, Turb, NO₂-N, and As; and for S4, COD, NO₂-N, BOD₅, Cond, TKN, Turb, and DO. The distinct parameter combinations identified for the global region and the four

subregions highlight the necessity of spatially adaptive regionalization in predicting river water quality.

3.2. Performance evaluation of water quality prediction models

3.2.1. Performance analysis of global models under different window lengths

In the 3-month sliding-window test, the global TCN-LSTM model demonstrates the best prediction performance among the four models, achieving the highest R^2 value of 0.724 and the lowest error metrics, with RMSE, MAE, and MAPE values of 5.033, 3.566, and 4.752%, respectively (Table 1). The next best-performing model is Transformer-ED, while LSTM and TCN present comparatively inferior results. In tests with sliding window of 6, 9, and 12 months, the Transformer-ED consistently outperforms the other global models, showing both the highest R^2 and the lowest RMSE, MAE and MAPE values. The TCN-LSTM model ranks second in prediction performance, whereas the LSTM and TCN models exhibit relatively weaker prediction capabilities.

Table 1

Performance comparison of global prediction models with window lengths of 3, 6, 9, and 12 months.

Sliding-window lengths	Deep learning algorithms	Performance metrics			
		R^2	RMSE	MAE	MAPE
3 months	LSTM	0.684	5.387	4.175	5.242%
	TCN	0.652	5.643	4.495	5.777%
	TCN-LSTM	0.724	5.033	3.566	4.752%
	Transformer-ED	0.715	5.117	4.203	5.193%
6 months	LSTM	0.754	4.747	3.613	4.701%
	TCN	0.744	4.846	3.975	4.996%
	TCN-LSTM	0.780	4.494	3.380	4.244%
	Transformer-ED	0.794	4.344	3.431	4.168%
9 months	LSTM	0.739	4.962	4.109	5.054%
	TCN	0.730	5.044	4.115	5.024%
	TCN-LSTM	0.755	4.814	3.821	4.747%
	Transformer-ED	0.779	4.572	3.612	4.639%
12 months	LSTM	0.724	5.096	4.047	4.882%

TCN	0.711	5.216	4.265	5.159%
TCN-LSTM	0.745	4.900	3.860	4.988%
Transformer-ED	0.768	4.668	3.683	4.622%

In the context of global modeling, the prediction performance of the TCN-LSTM and Transformer-ED models consistently surpasses that of the TCN and LSTM models. This suggests that, for global water quality prediction, models employing the encoder-decoder architecture outperform those based on conventional single-network structures, which highlights the advantages of encoder-decoder frameworks in handling complex time series water quality data. Although the optimal global water quality prediction models have been identified under different window lengths, the performance metrics reveal that the overall prediction accuracy of these models remains suboptimal. This finding suggests that further optimization of prediction models is necessary, particularly to enhance the regional adaptability of the prediction models by considering the spatial heterogeneity of water quality.

3.2.2. Performance analysis of regional models under different window lengths

(1) The tests with 3-month sliding-window length

Building upon the results of the spatially adaptive regionalization, the regional models for water quality prediction are developed within the four subregions using four deep learning algorithms. When the window length is set to 3 months, test results for subregion 1 (S1) indicates that the LSTM model achieved the best prediction performance, with the highest R^2 (0.63), the lowest RMSE (4.19), and the lowest MAE (3.38) (Fig. 4). The TCN model ranks second, while TCN-LSTM and Transformer-ED exhibits relatively poorer prediction performance. In subregions 2 (S2) and 3 (S3), the TCN model outperforms the other models, followed by LSTM, whereas TCN-LSTM and Transformer-ED again shows the least satisfactory results. In subregion 4 (S4), based on the performance metrics, the order of model performance from highest to lowest is LSTM, TCN, TCN-LSTM, and Transformer-ED. Therefore, it can be concluded that under the 3-month sliding window, the regionally optimal models for the four subregions are LSTM (S1), TCN (S2), TCN (S3), and LSTM (S4).

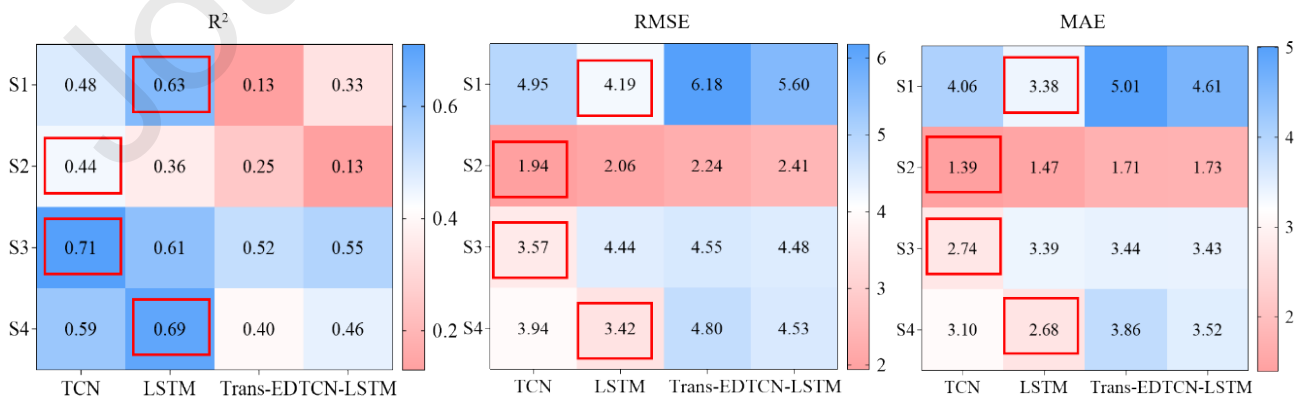


Fig. 4. Performance metrics of regional prediction models for a 3-month window length.

(2) The tests with 6-month sliding-window length

When the sliding-window length is set to 6 months, test results for S1, S3, and S4 indicate that the LSTM model demonstrates the strongest prediction capability, with the highest R^2 and the lowest RMSE and MAE (Fig. 5). In these subregions, the TCN model takes the second position in prediction performance, while the TCN-LSTM and Transformer-ED models perform relatively poorly. In subregion 2, the TCN model achieves the highest R^2 (0.46) and the lowest RMSE (1.73), suggesting its superior performance in both variance explanation and prediction accuracy, particularly in capturing the overall trend of water quality fluctuations. The LSTM model, on the other hand, obtains the lowest MAE (1.29), indicating it is more effective in handling individual outliers than TCN. These differences in performance metrics reveal the distinct advantages of the TCN and LSTM models. Considering the overall characteristics of the water quality data, TCN provides a more balanced and reliable prediction performance. Therefore, under the 6-month sliding window, the optimal models for the four subregions are identified as LSTM (S1), TCN (S2), LSTM (S3), and LSTM (S4).

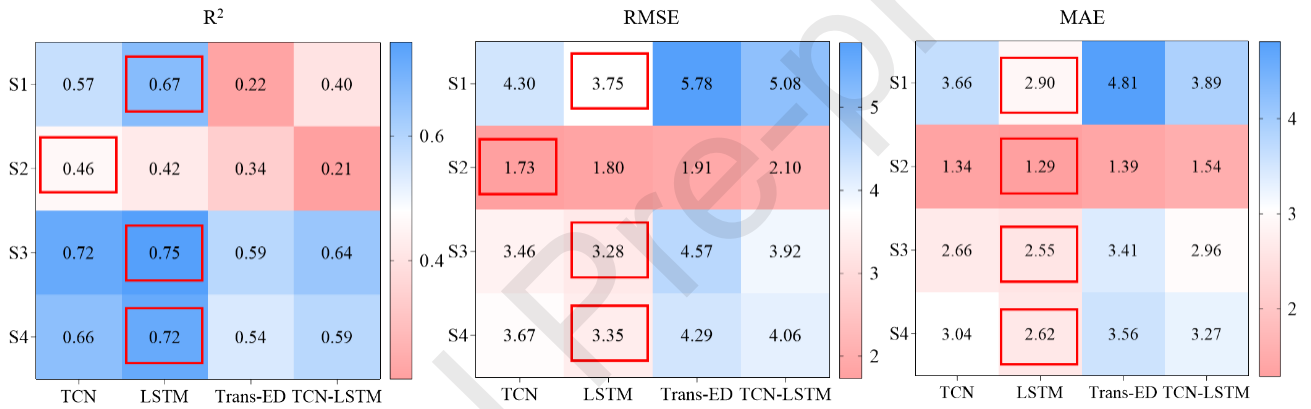


Fig. 5. Performance metrics of regional prediction models for a 6-month window length.

(3) The tests with 9-month sliding-window length

When the sliding-window length is set to 9 months, by comparing the performance metrics (R^2 , RMSE, and MAE), it can be known that the optimal models for S1, S3, and S4 are LSTM. For the other models, the prediction performance ranks in descending order as TCN, Transformer-ED, and TCN-LSTM (Fig. 6). Notably, in S2, although the LSTM model attains the highest R^2 value, the difference compared to the TCN model is marginal, suggesting that LSTM has a slight advantage in explaining data variance. However, the TCN model obtains the lowest RMSE and MAE values, indicating superior prediction accuracy and better handling of outliers. Considering the magnitude of the differences in performance metrics among the four models, the TCN model is ultimately selected as the optimal model for S2. Accordingly, at a sliding-window length of 9 months, the optimal models for the subregions are identified as LSTM (S1), TCN (S2), LSTM (S3), and LSTM (S4).

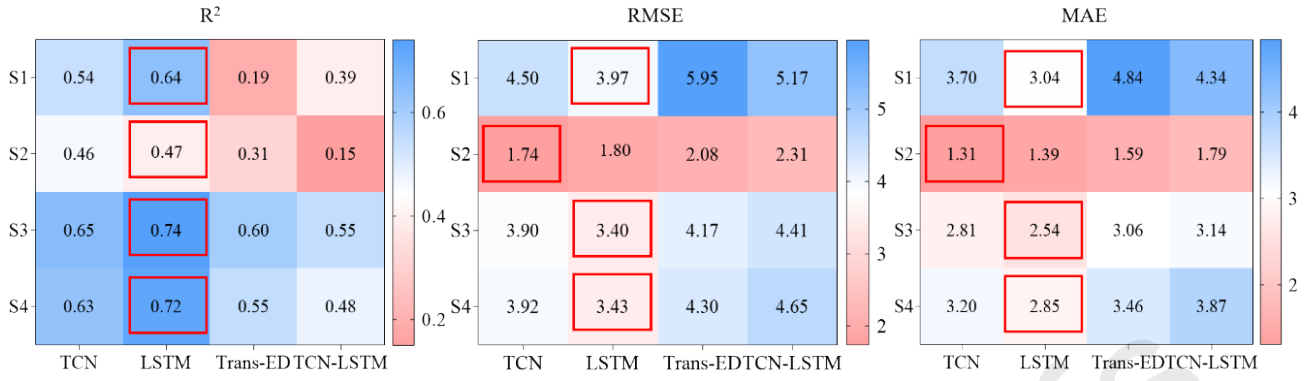


Fig. 6. Performance metrics of regional prediction models for a 9-month window length.

(4) The tests with 12-month sliding-window length

When the sliding-window length is set to 12 months, the test results for S1, S2, and S4 indicate that the LSTM model exhibits the strongest water quality prediction performance. The TCN model ranked second, whereas the TCN-LSTM and Transformer models demonstrate comparatively weaker capabilities (Fig. 7). In S3, the TCN model achieves the highest R² (0.72) and the lowest RMSE (3.39) and MAE (2.69), reflecting superior performance in terms of variance explanation, prediction accuracy, and robustness to outliers. Therefore, under the 12-month window length, the regionally optimal models for the four subregions are identified as LSTM (S1), LSTM (S2), TCN (S3), and LSTM (S4).

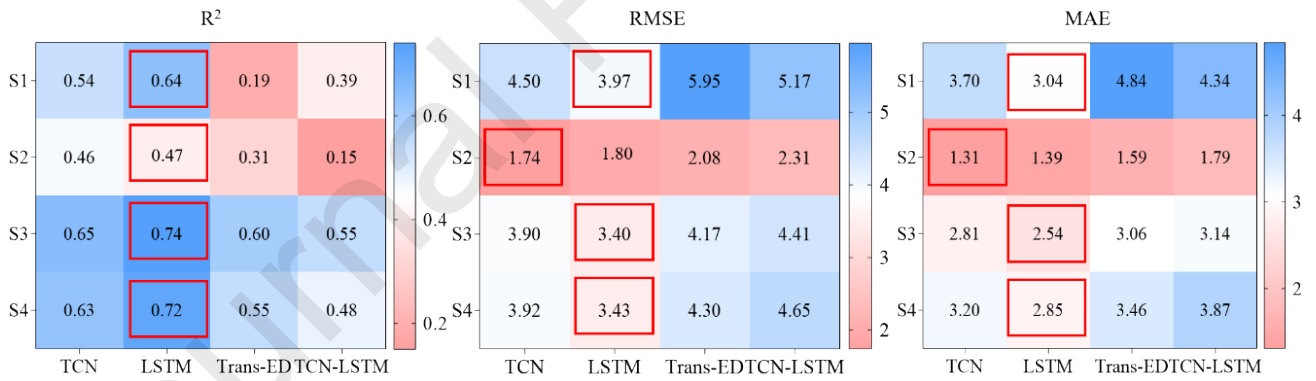


Fig. 7. Performance metrics of regional prediction models for a 12-month window length.

3.2.3. Performance comparison

To analyze differences in the performance of global and regional prediction models, the best-performing models are identified under different sliding-window lengths. For the global prediction task, the TCN-LSTM with a 3-month window and the Transformer-ED with 6-month, 9-month, and 12-month windows attain the highest prediction performance. For the regional prediction task, the optimal models vary across subregions (S1-S4) depending on the window length. Under the 3-month window, the optimal model is LSTM (S1)-TCN (S2)-TCN (S3)-LSTM (S4). For the 6-month and 9-month windows, the same optimal configuration is

observed: LSTM (S1)-TCN (S2)-LSTM (S3)-LSTM (S4). Under the 12-month window, the optimal model is LSTM (S1)-LSTM (S2)-TCN (S3)-LSTM (S4).

As shown in Fig. 8 (Comparison 1), the regional models consistently achieve lower error metrics (RMSE, MAE, and MAPE) than the global models across different sliding-window lengths. Meanwhile, the regional models also surpass the global models in terms of R^2 and CC metrics. Compared to the global models, regional models with window lengths of 3, 6, 9, and 12 months significantly improve performance. R^2 increases by 22.38%, 13.73%, 15.15%, and 16.15%, and CC by 6.08%, 4.74%, 2.82%, and 6.91%. Meanwhile, RMSE decreases by 35.86%, 31.19%, 31.87%, and 31.49%, MAE by 32.61%, 33.98%, 35.96%, and 33.07%, and MAPE by 36.62%, 32.34%, 37.12%, and 33.28%, respectively. These results indicate that the regional models not only yield smaller prediction errors but also a higher percentage of explained variance and stronger correlation between predicted and actual values. Similarly, the scatter plots in Fig. 8 (Comparison 2) further confirm the superiority of the regional models. Relative to the global models, the yellow scatter points of the regional models are more tightly clustered around the diagonal across all window lengths, and their regression line slopes are closer to 1. This pattern demonstrates that the regional models capture regional variations more effectively, producing WQI predictions that align more closely with actual values. The line charts in Fig. 8 (Comparison 3) present the residuals of global and regional water quality predictions. The errors of the regional models (in red) are consistently smaller and more concentrated than those of the global models (in blue) in all window lengths, reflecting the greater stability and robustness of regional prediction models. Taken together, this performance difference can be attributed to the regional models' superior ability to capture localized patterns and trends in the data, whereas the global models are less sensitive to such localized variations. Consequently, regional division-based modeling has a clear advantage over global modeling for water quality prediction, which provides more accurate and reliable results.

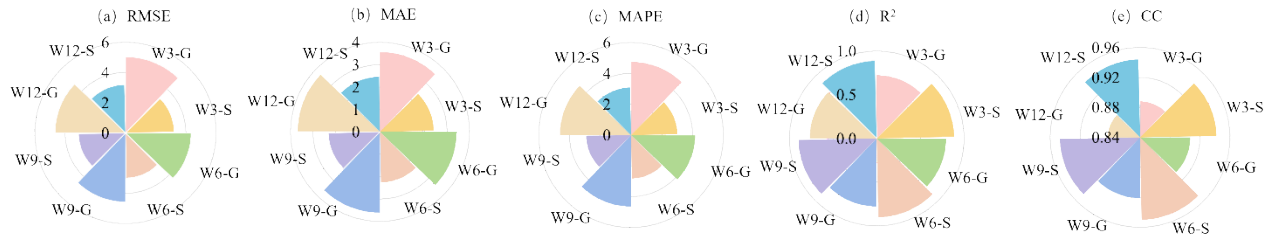
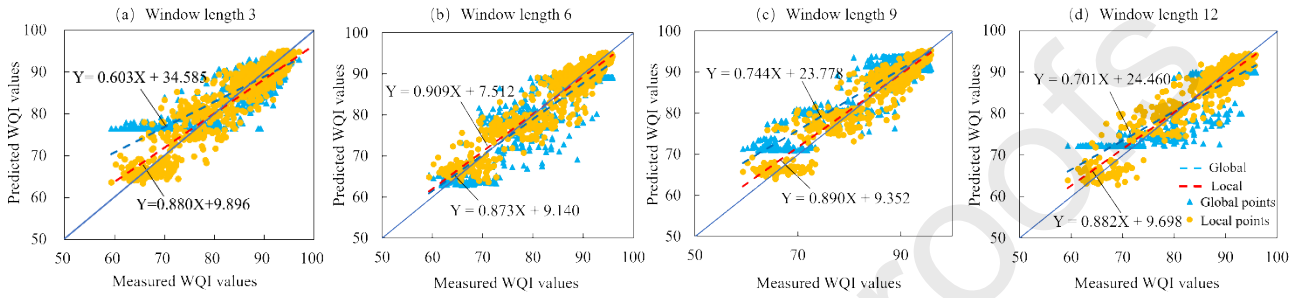
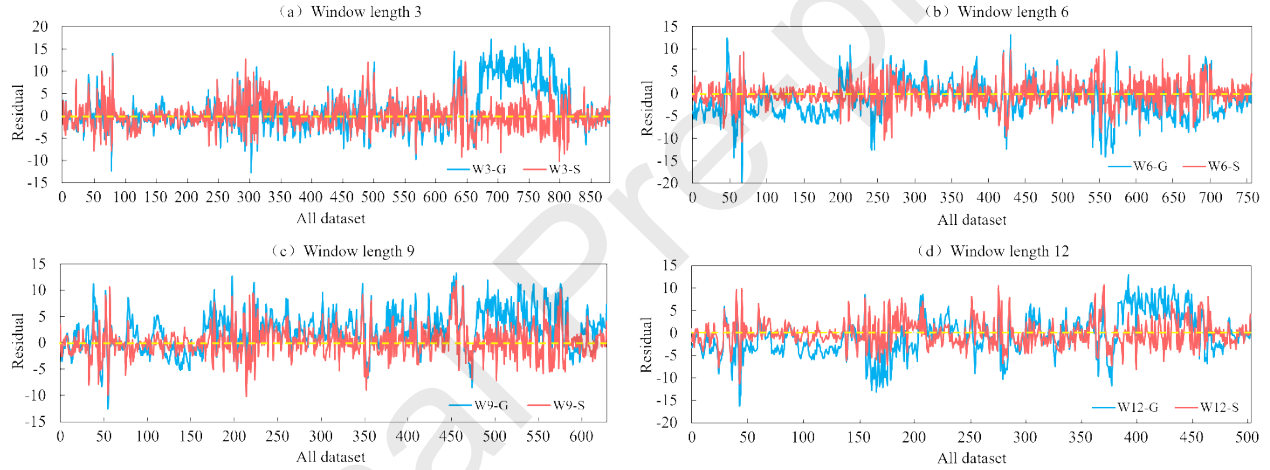
Comparison 1 Performance metrics analysis**Comparison 2 Regression analysis****Comparison 3 Residual analysis**

Fig. 8. Performance comparison between global and regional prediction models under different window lengths (S represents the subregion prediction model, G represents the global prediction model).

3.3. Identification of the optimal model for water quality prediction

Based on the foregoing analysis, the optimal prediction model for the study area is identified among the regional prediction models. As illustrated in Comparison 1 of Fig. 8, the regional prediction model with a 6-month window length exhibits RMSE, MAE, and MAPE error metrics clustered nearer to the center of the plot, while the fitting metric R^2 and correlation coefficient (CC) are positioned closer to the periphery. The pattern indicates that adopting a 6-month window length leads to water quality prediction with lower and less dispersed errors, together with superior goodness-of-fit and stronger correlation. The model with a 9-month window length performs slightly less well, followed by the 12-month window model, while the model with a 3-month window length shows the poorest performance across all performance metrics. According to Comparison 2 in Fig. 8, the regional prediction model with a 6-month sliding window also presents a regression line slope closest to 1, and yellow scatter

points are more densely distributed along both sides of the diagonal, signifying the best agreement between predicted values and actual values. As the window length increases or decreases, the regression slopes decrease accordingly, ranked in descending order for the 9-, 12-, and 3-month windows. Among these, the 9-month models have more yellow scatter points distributed around the diagonal, followed by the 12-month model, whereas the 3-month model shows a more dispersed distribution. This suggests that the differences between the predicted and actual values increase in the order of 9, 12, and 3 months. In Fig. 8 (Comparison 3), the regional model with a 6-month window length exhibits the smallest residual fluctuations between the predicted and actual values, indicating its prediction results are not only closer to the actual values but also more stable. Considering Comparisons 1, 2, and 3, the regional model with a 6-month window (LSTM (S1)-TCN (S2)-LSTM (S3)-LSTM (S4)) demonstrates the best overall performance. The 9-month and 12-month models perform less well, while the 3-month model exhibits the weakest prediction performance.

3.4. Performance interpretation of the optimal model

The SHAP method, an explainable artificial intelligence approach, is applied to interpret model behavior at both global and regional levels. It computes the marginal contribution (SHAP value) of each feature to the prediction outcome (Baptista et al., 2022; Zhang et al., 2023; Yao et al., 2024; Wang et al., 2022). A key advantage of SHAP is its ability to quantify the influence of each feature for individual samples, while also indicating the direction of the effect (positive or negative). The input feature parameters of the optimal model under a 6-month window length are analyzed. Fig. 9 to Fig. 12 illustrate the effects and relative importance of eight feature parameters on the predicted WQI values.

3.4.1. Contribution of feature parameters in the regionally optimal model for subregion 1

Fig. 9a illustrates the mean absolute SHAP values for different feature parameters, reflecting their average contribution to the model's prediction outcomes. Longer bars represent feature parameters with greater average importance. Among the input features, historical WQI exerts the largest influence on the model, followed by F^- , and then DO (Fig. 9(a)). The remaining feature parameters, ranked in descending order of their impact on the predicted WQI values, are BOD_5 , NO_2-N , COD, SS, and As. In Fig. 9b, each dot represents an individual sample from the dataset, and its position corresponds to the value indicating the specific contribution of the input feature to the model prediction. The historical WQI exhibits a positive effect on the predicted WQI—namely, higher input WQI values are associated with higher predicted outputs. However, the historical concentration of F^- in the water presents a negative impact on the predicted WQI values, indicating that an increase in F^- concentration tends to lower future WQI levels. Similarly, DO also shows a negative influence on the predicted WQI, suggesting that higher DO concentrations may lead to a decrease in WQI predictions. This finding contrasts with previous studies and may be attributed to the fact that, once the DO concentration reaches a level adequate to sustain aquatic life and ecosystem balance, further increases in DO are likely to trigger excessive algal growth, resulting in eutrophication-related water quality issues. In addition, the remaining five feature parameters (BOD_5 , NO_2-N , COD, SS, and As) also exhibit negative effects on the predicted WQI, implying that increases in their

concentrations are associated with declines in future water quality.

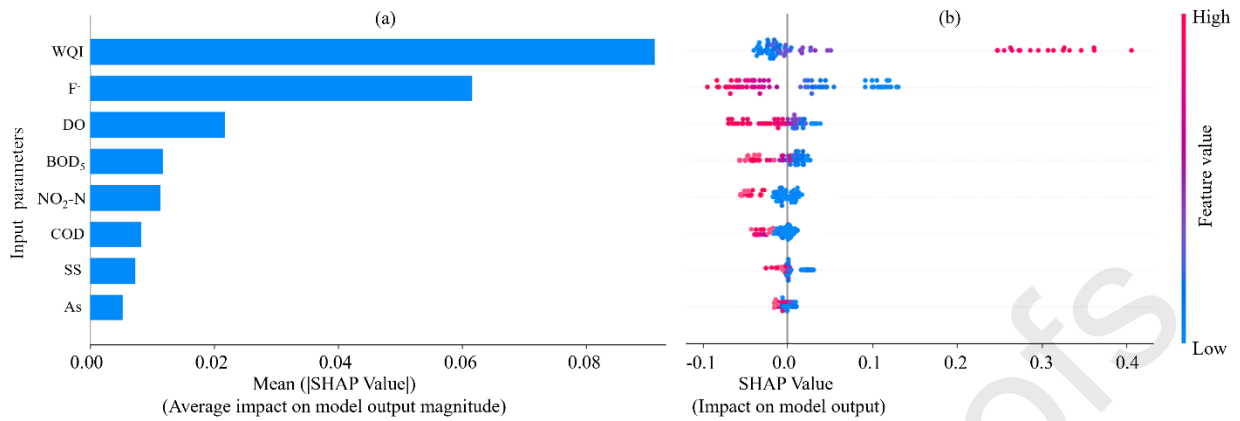


Fig. 9. Impact of feature parameters on output WQI in S1: (a) distribution of mean absolute SHAP values; (b) distribution of SHAP values.

3.4.2. Contribution of feature parameters in the regionally optimal model for subregion 2

As shown in Fig. 10a, the historical WQI exhibits the highest mean absolute SHAP values among these features, indicating that the historical state of WQI has the most significant influence on future WQI variations. The subsequent input parameters are ranked in order of importance as follows: NH₄-N, F⁻, and Cond. The remaining four parameters contribute to the model performance in descending order: COD, NO₂-N, SS, and BOD₅. Fig. 10b further demonstrates the historical WQI displays a positive effect on the predicted WQI, while all other input features show negative effects. This pattern is consistent with findings reported in multiple previous studies (Wang et al., 2022; Varol and Tokatlı, 2023; Simeonov et al., 2003). NH₄-N, NO₂-N, and F⁻ serve as indicators of environmental pressure, and their elevated concentrations are naturally associated with water quality deterioration. Increased NH₄-N concentrations can promote eutrophication, stimulating excessive algal growth, depleting DO, and disrupting aquatic ecosystems. Elevated NO₂-N levels may enhance water toxicity, directly harming aquatic organisms; moreover, nitrite can be converted into carcinogenic compounds, posing risks to human health and compromising water quality safety. Excessive concentrations of F⁻ can result in fluoride contamination, exerting toxic effects on aquatic life and disrupting ecological balance. In addition, increased Cond reflects a higher concentration of dissolved mineral salts, which can lead to water hardness and adversely affect aquatic ecosystems. Elevated SS levels reduce water transparency, limit photosynthesis, and impose stress on the survival of aquatic organisms. Rising concentrations of COD and BOD₅ indicate organic pollution, which diminishes the self-purification capacity of the water body, destabilizes ecological stability, and threatens water quality safety.

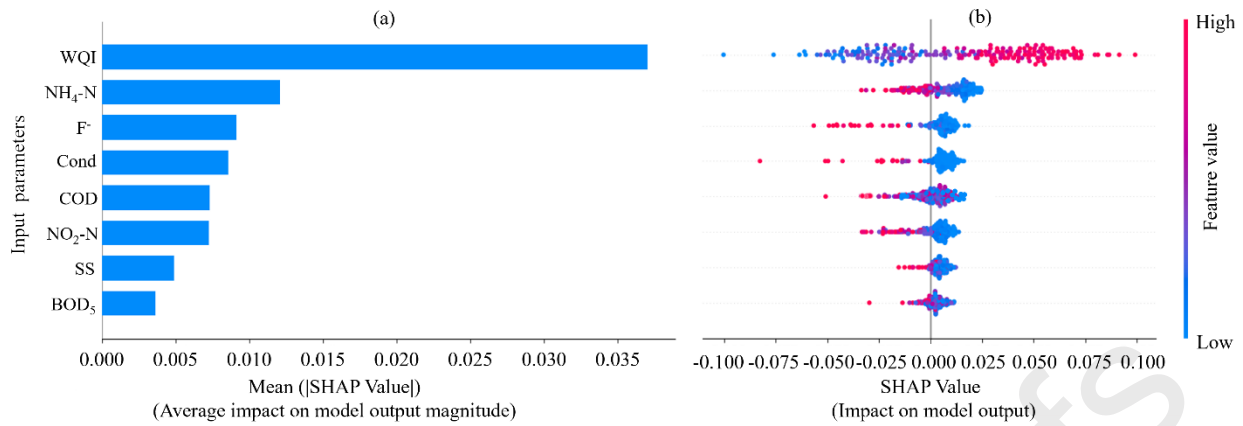


Fig. 10. Impact of feature parameters on output WQI in S2: (a) distribution of mean absolute SHAP values; (b) distribution of SHAP values.

3.4.3. Contribution of feature parameters in the regionally optimal model for subregion 3

In Fig. 11a, the historical WQI also displays the most significant impact on the predicted results, followed by Cond, TKN, and F⁻. The remaining input parameters are ranked in descending order of influence as As, Cu, Turb, and NO₂-N. As illustrated in Fig. 11b, the historical WQI exerts a positive impact on the predicted outcomes, whereas all other features exhibit negative effects. The influences of Cond, F⁻, and NO₂-N have been discussed previously; here, the analysis focuses on the impacts of TKN, As, Cu, and Turb. TKN, representing the total concentration of organic nitrogen and ammonia nitrogen in the water body, may lead to eutrophication when elevated. This can stimulate excessive algal growth, which in turn overconsumes DO and deteriorates water quality, ultimately disrupting aquatic ecosystems (Su et al., 2019). As, a toxic metalloid, poses risks to both aquatic organisms and human health, and its negative impact on future WQI values is consistent with previous findings (Cüce et al., 2022; Njuguna et al., 2020). Although Cu is an essential trace element for many organisms, excessive concentrations can exert toxic effects on aquatic environments and organisms, thus impairing water quality. Increased Turb reflects a rise in suspended particulate matter, which may carry pathogens and pollutants, thus contributing to the degradation of future water quality.

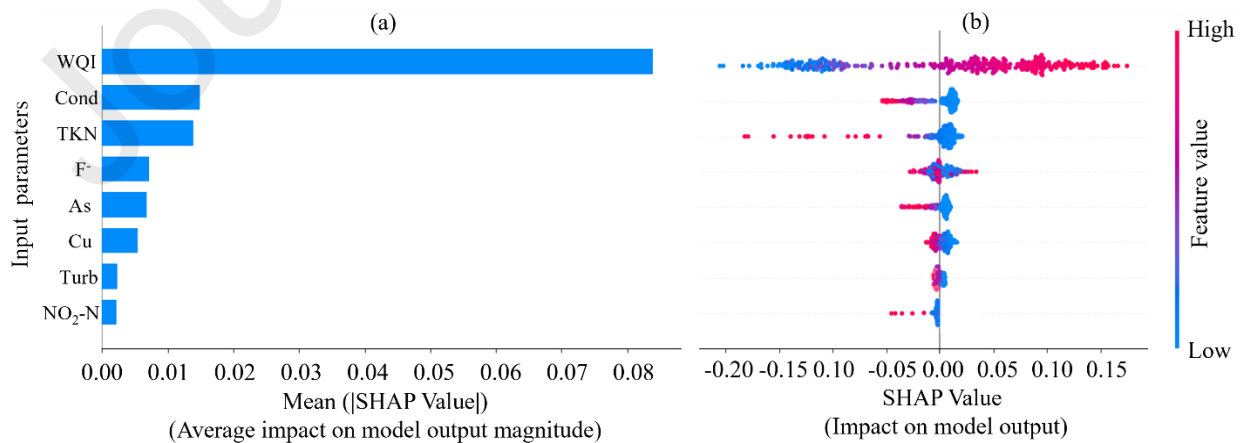


Fig. 11. Impact of feature parameters on output WQI in S3: (a) distribution of mean absolute SHAP

values; (b) distribution of SHAP values.

3.4.4. Contribution of feature parameters in the regionally optimal model for subregion 4

Fig. 12a depicts the influence of eight feature parameters on the predicted WQI. Among them, the historical WQI still exerts the most significant effect, followed by Cond and TKN. The remaining five parameters, in descending order of importance, are DO, NO₂-N, COD, BOD₅, and Turb. As shown in Fig. 12b, the historical WQI and DO positively contribute to the predicted results, whereas the other six parameters have negative effects. Notably, the impact of DO in this subregion contrasts with its role in S1. DO is one of the key indicators of aquatic ecosystem health, serving a crucial function in sustaining the physiological functions of various aquatic species. Moreover, it contributes significantly to the microbial degradation of organic matter, which helps to lower the concentration of potentially harmful substances and to mitigate their adverse effects on the ecosystem. When a water body exhibits strong self-purification capacity, it can efficiently process and neutralize incoming pollutants, a capacity typically associated with adequate DO concentrations. Therefore, within a certain range, an increase in DO concentration often indicates water quality improvement, which explains the positive relationship between DO and WQI in this subregion. The effects of the remaining seven parameters on WQI predictions are consistent with the patterns observed in other subregions.

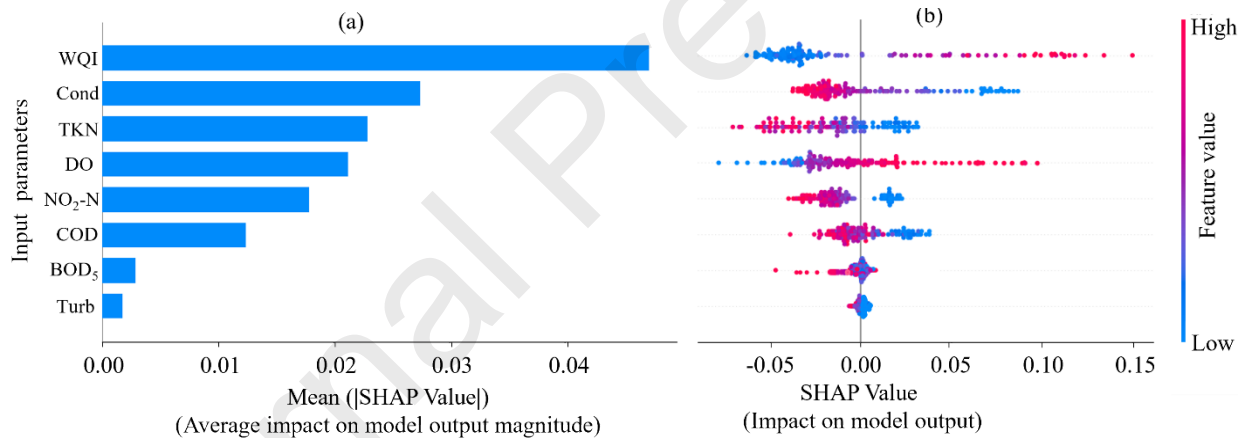


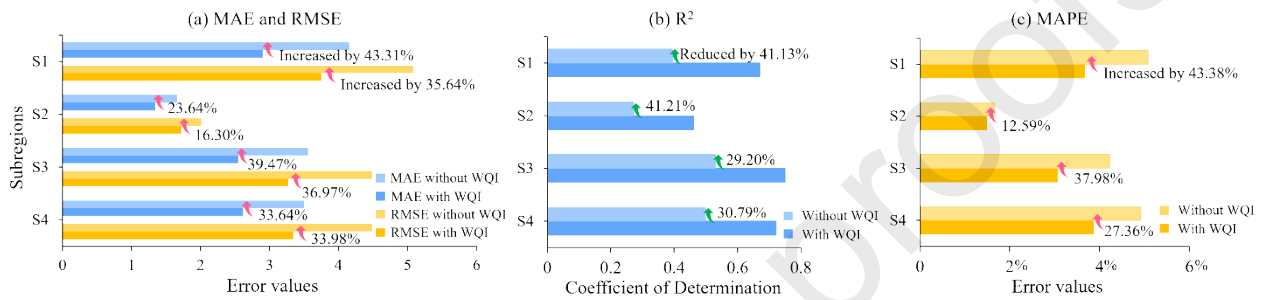
Fig. 12. Impact of feature parameters on output WQI in S4: (a) distribution of mean absolute SHAP values; (b) distribution of SHAP values.

3.5. Quantitative influence of historical WQI on prediction performance

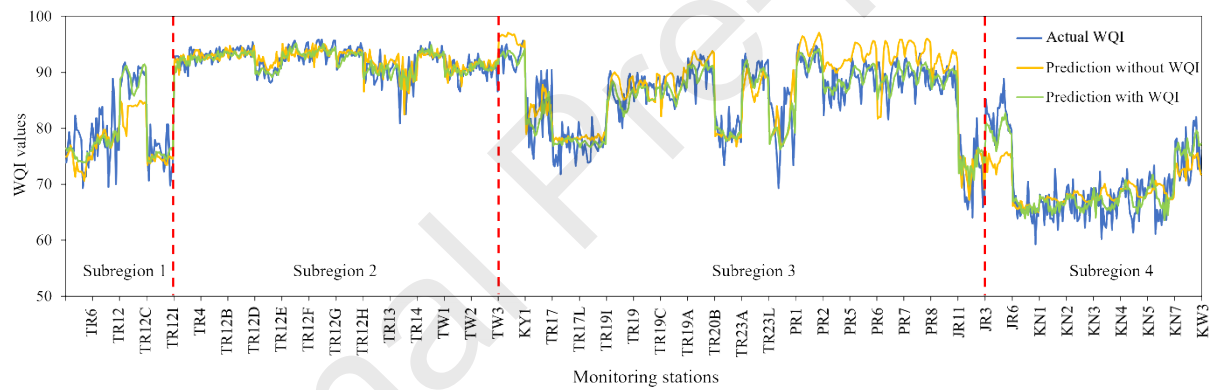
To quantitatively evaluate the impact of historical WQI on water quality prediction, the prediction performance of the regionally optimal models is analyzed by varying whether historical WQI is included in the feature parameter. As presented in Fig. 13 (Comparison 1), models incorporating historical WQI as input demonstrate a significant improvement in prediction performance compared to those without historical WQI. This effect is particularly pronounced in S1, S3, and S4, where LSTM models with a 6-month window length are adopted. In these subregions, the models excluding historical WQI as input parameter exhibit increases of at least 33.64%, 33.98%, and 27.36% in MAE, RMSE, and MAPE respectively, and a decrease of at least 29.20% in R², compared to models including historical WQI. For S2, a

TCN model with a 6-month window length is utilized to predict water quality. In this subregion, although the changes in error metrics for the prediction model are relatively smaller than in the other subregions, the difference remained substantial. When historical WQI is considered as input, MAE, RMSE, and MAPE increased by 23.64%, 16.30%, and 12.59%, respectively, while R^2 decreased by 41.21%. These performance differences indicate that the historical WQI values play a critical explanatory role for its future WQI variations. Incorporating historical WQI as an input allows the model to leverage past temporal dependencies, improving prediction accuracy, whereas models without it struggle to adequately capture complex relationships between other features and future WQI.

Comparison 1 Performance variation analysis



Comparison 2 Prediction accuracy analysis



Comparison 3 Regression analysis

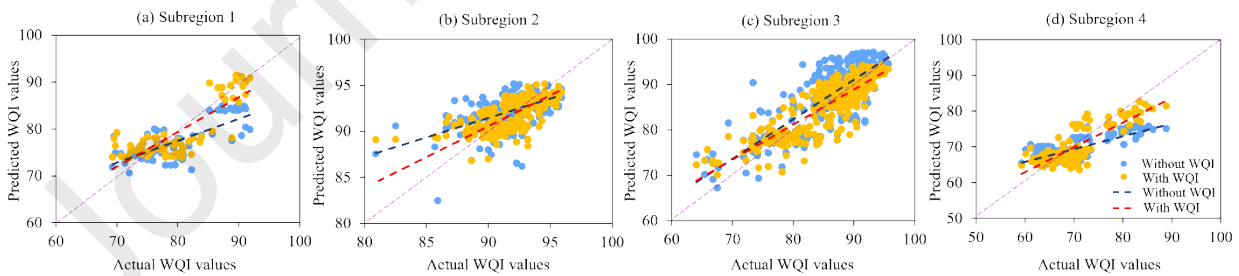


Fig. 13. Performance comparison of prediction models with and without historical WQI as an input feature.

Fig. 13 (Comparison 2) clearly illustrates the performance differences in study area when predicting WQI using the regionally optimal models with or without historical WQI as an input feature. The green line (predictions with historical WQI) closely follows the blue line (actual values), suggesting that models incorporating historical WQI can accurately capture both the trend and magnitude of WQI fluctuations in the dataset. These models demonstrate neither

overfitting nor underfitting, underscoring their robust performance in water quality prediction. In contrast, the yellow line (predictions without historical WQI) deviates noticeably from the blue line, reflecting a significant decline in prediction accuracy when historical WQI is excluded from the model input. Fig. 13 (Comparison 3) reveals that prediction accuracy varies across subregions, which can be attributed to the significant spatial heterogeneity of water quality features in different areas. Moreover, in all four subregions, it is consistently observed that the red regression line is closer to the purple diagonal than the blue regression line, and the yellow dots are more tightly clustered around the diagonal compared to the blue ones. These results underscore that the inclusion of historical WQI as an input feature substantially enhances the ability of prediction models to capture the evolving patterns of water quality.

4. Discussion

4.1. Analysis and interpretation of the findings

In global prediction scenarios, TCN-LSTM and Transformer-ED outperform TCN and LSTM, but for regional predictions, simpler models like TCN and LSTM achieve better performance. The results indicate that complex encoder-decoder models can effectively leverage large, heterogeneous datasets for global predictions, but simpler architectures may provide more robust performance when training data are limited or localized. This finding is consistent with previous research indicating that encoder-decoder models, such as Transformer-ED, are well-suited for capturing complex, long-range dependencies in time-series data (Wang et al., 2023; Lin et al., 2024; Bi et al., 2025). Specifically, Transformer-ED's self-attention mechanism excels in global modeling scenarios where extensive datasets are available (Hu et al., 2024; Huan et al., 2025). Similarly, the TCN-LSTM hybrid, by combining TCN's ability to capture long-range dependencies with LSTM's sequential memory, has demonstrated superior performance in various complex forecasting tasks (Bi et al., 2021; Xie et al., 2024; Li et al., 2022). However, when transitioning to regional prediction tasks, the results of this study differ. Simpler models like LSTM and TCN perform better in regional predictions, which is consistent with findings from Sonata and Heryadi (2024), who observed that simpler models LSTM showed higher performance for the time series data forecasting process compared with Transformer. This is likely due to the overfitting risk associated with more complex architectures, which may not generalize well to the smaller, region-specific datasets. Our study also supports the notion that LSTM's ability to capture long-term dependencies and TCN's causal convolutions make them ideal for regional models where the data may exhibit less complexity. This comparison highlights that increasing model complexity does not always improve prediction accuracy and that the most suitable model depends on the interplay between dataset size, regional heterogeneity, and the temporal dynamics of water quality.

The SHAP analysis revealed that historical WQI is the most critical input across all subregions, consistently exerting a strong positive effect on predictions. However, other water quality parameters usually show spatially heterogeneous contributions. For example, DO negatively influenced WQI predictions in subregion 1 but had a positive effect in subregion 4.

This diversity in effect could be attributed to variability in ecological and biogeochemical conditions. Adequate DO supports aquatic life and microbial degradation of organic matter, improving water quality, whereas excessive DO can stimulate algal growth and eutrophication, temporarily reducing water quality. Further investigation combining SHAP analysis with field measurements and process-based modeling is needed to more rigorously link these water quality drivers to underlying hydrological and biogeochemical processes.

4.2. Uncertainties associated with data and modeling framework

In this section, we discuss several sources of uncertainty that may affect the accuracy and reliability of water quality predictions. These uncertainties stem from the quality of monitoring data, the regionalization of water quality subregions, and the selection of prediction model parameters.

4.2.1. Uncertainty in water quality monitoring data

Water quality monitoring data are subject to several sources of uncertainty. These include measurement errors, sensor calibration issues, missing values, and general noise inherent in environmental data (Zhang and Thorburn, 2022; Li et al., 2024; Chen and Xue, 2023; Ngouna et al., 2020; Weber and Bieroza, 2026). Despite efforts to preprocess and clean the data, these factors can still introduce variability into the model's predictions, particularly in regions with sparse or inconsistent data coverage.

The input data uncertainty can influence subsequent analyses in two main ways. First, it may affect the spatial regionalization process, as clustering algorithms like REDCAP rely on observed water quality values to define relatively homogeneous subregions. Noise or missing data could slightly alter cluster boundaries or the assignment of monitoring stations, which in turn may impact the characteristics of the derived subregions. Second, uncertainty in input data can affect prediction performance. Key water quality parameters and historical WQI sequences were used as inputs for each subregion, and noise or missing values may lead to incomplete or biased features. Subregions with sparse or highly variable data may therefore show slightly higher prediction bias.

4.2.2. Uncertainty in water quality regionalization

Uncertainty in water quality regionalization arises primarily from the selection of the number of subregions and the definition of their boundaries. Different choices can lead to variations in which monitoring stations are assigned to each subregion and how local patterns are represented. In our experiments, multiple partition schemes were tested in this study, including 2 to 6 subregions. The four-subregion scheme was selected based on the lowest sum of squared deviations, balancing within-region homogeneity and overall data representativeness. This ensures that the derived subregions are both environmentally meaningful and geographically consistent. Despite these measures, uncertainty in regionalization can still affect prediction performance. Specifically, slight changes in subregion boundaries or composition may alter the local data features that serve as inputs to the prediction models. Such variability can propagate through the models, potentially affecting the accuracy

of predicted water quality metrics at the subregion level.

4.2.3. Uncertainty in water quality prediction model parameters

Uncertainty in prediction model parameters arises mainly from the selection of sliding-window length and the tuning of hyperparameters such as learning rate, batch size, and number of iterations.

The sliding-window length determines how much historical information is included in each prediction, influencing the model's ability to capture both short-term fluctuations and long-term trends. Too short a window may fail to capture sufficient temporal patterns, whereas too long a window may incorporate excessive seasonal cycles or noise. In this study, sliding windows of 3, 6, 9, and 12 months were tested by evaluating the performance of both regional and global models. The results demonstrated that the 6-month window consistently yielded lower and less dispersed prediction errors, with higher goodness-of-fit and correlation compared to the other window lengths. It should be noted that this optimal window length may be specific to our study area. Different regions may exhibit distinct climate patterns, hydrological responses, or land use characteristics, which influence the temporal dynamics of water quality. In such cases, a different window length (e.g., 12 months) may be necessary to adequately capture local seasonal trends and short-term fluctuations.

Hyperparameter tuning can also introduce uncertainty that may affect the model's ability to capture temporal patterns. In this study, the training dataset (2010-2017) was randomly partitioned into 5 mutually exclusive folds for cross-validation, while the later period (2018-2019) was reserved as an untouched test set. Importantly, the temporal order within each sample, represented by historical sequences of water quality parameters, was preserved when constructing input features. This ensures that predictions for each time step are based only on past observations, maintaining the temporal dependence of the data. Sensitivity analyses using alternative window lengths and cross-validation folds confirmed that the selected hyperparameters produced stable and reliable predictions across both regional and global models. These procedures help mitigate the impact of hyperparameter uncertainty while preserving the integrity of temporal information in water quality prediction.

4.3. Limitations and future perspectives

Despite the promising performance of the proposed regional water quality prediction framework, several limitations remain that provide avenues for future research. First, the uncertainty inherent in input data, regionalization, and model parameters can still affect prediction reliability. Future work should explore additional robustness checks, such as expanded cross-validation schemes and systematic sensitivity analyses, to further quantify and reduce the impact of these uncertainties. Incorporating probabilistic prediction frameworks or uncertainty quantification methods may also enhance model credibility and provide more informative outputs for water quality management.

Second, generalizing the model to other regions with different climate, topography, and land use requires careful consideration. The optimal sliding-window length for temporal

feature extraction may vary across regions due to differences in seasonal and inter-annual patterns. In future work, techniques inspired by multi-scale time series analysis, such as wavelet-based methods (Ghaderpour et al., 2024), could be applied to guide the selection of window length and to better capture both short- and long-term temporal components. This approach would enhance the transferability and robustness of water quality prediction models under diverse environmental conditions.

5. Conclusion

This study proposes a spatial-heterogeneity-aware framework to improve river water quality prediction under spatially heterogeneous conditions. Four relatively homogeneous subregions are delineated using spatially adaptive clustering, and subregion-specific key WQPs are identified. Both global and regional models are constructed with four deep learning algorithms and evaluated under varying sliding-window lengths to identify the optimal model. The SHAP method is then applied to explain the influence of key input parameters on WQI predictions. A case study in Hong Kong demonstrates the following findings:

(1) In global modeling, TCN-LSTM and Transformer-ED consistently outperform TCN and LSTM across all sliding windows, whereas in regional modeling, LSTM and TCN perform better than TCN-LSTM and Transformer-ED.

(2) Regional models consistently outperform global models under 3-, 6-, 9-, and 12-month sliding windows, achieving at least a 13.73% improvement in R^2 and reductions of at least 31.19%, 32.61%, and 32.34% in RMSE, MAE, and MAPE.

(3) Sliding-window length markedly affects prediction performance, and the optimal model is a combination of four regional models with a 6-month window, configured as LSTM (S1)-TCN (S2)-LSTM (S3)-LSTM (S4).

(4) Historical WQI is the most critical input across all subregions with a strong positive effect on predictions. Other WQPs, such as F^- , NH_4-N , NO_2-N , and DO, exhibit contributions that vary among subregions, highlighting the spatially heterogeneous roles of drivers of water quality.

As discussed in Section 4.3, future work will further enhance model reliability and generalizability by quantifying uncertainties and adapting the framework to regions with diverse environmental conditions.

Author's Contributions

Qiaoli Wang: Writing—original draft, Writing—review & editing, Software, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. Min Deng: Writing—review & editing, Funding acquisition. Bingbing Zhao: Software, Visualization. Jiannan Cai: Writing—

review & editing, Conceptualization, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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